Notes on the Likelihood Function

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Abstract

Notes developing the likelihood function from the binomial example: the shape of $L(q \mid v, n)$, the sense in which the likelihood "is" the probability mass function with argument and parameter exchanged, and the measure-theoretic definition together with the three qualifications it carries.

Contents

1 The binomial example and the shape of the likelihood

Consider the binomial model with probability of success q and n known. The probability mass function of the count Υ is

$$f_\Upsilon(v; q, n) = \mathbb{P}(\Upsilon = v \mid n, q) = \binom{n}{v} q^v (1-q)^{n-v}, \qquad v \in \{0, 1, \ldots, n\}. \tag{1}$$

Fixing an observed count v and reading (1) as a function of q gives the likelihood

$$L(q \mid v, n) = \binom{n}{v} q^v (1-q)^{n-v}, \qquad q \in [0, 1]. \tag{2}$$

Why it has that shape. As a function of q, expression (2) is the kernel of a $\mathrm{Beta}(v+1,\, n-v+1)$ density; everything about its shape follows from that. The binomial coefficient is constant in q: it rescales vertically and plays no role in location or shape. The shape is the product of two competing power terms — q^v rewards larger q (pulling mass toward 1), while $(1-q)^{n-v}$ rewards smaller q (pulling toward 0) — and they balance at an interior point.

Work with the log-likelihood

$$\ell(q) = \text{const} + v \log q + (n-v) \log(1-q). \tag{3}$$

Then

$$\ell'(q) = \frac{v}{q} - \frac{n-v}{1-q} = 0 \implies v(1-q) = (n-v)q \implies q = \frac{v}{n}, \tag{4}$$

which is exactly the MLE, and

$$\ell''(q) = -\frac{v}{q^2} - \frac{n-v}{(1-q)^2} < 0 \quad \text{on } (0,1), \tag{5}$$

so ℓ is strictly concave. Hence L is strictly log-concave, therefore unimodal with a single peak at $q = v/n$.

For $0 < v < n$ the endpoints vanish, $L(0) = L(1) = 0$, giving the characteristic single bump: rising from 0, peaking at v/n, decaying back to 0. The degenerate cases collapse the bump against a wall: $v = 0$ gives $(1-q)^n$, monotone decreasing with mode at $q = 0$; $v = n$ gives q^n, monotone increasing with mode at $q = 1$.

Remark. L is smooth (a degree-n polynomial in q), which is what is meant by calling it "continuous." It is *not* a density in q: it does not integrate to 1 over q. Normalizing requires dividing by $\mathrm{B}(v+1, n-v+1)$, and the fact that this normalization produces the Beta distribution is precisely Beta–Binomial conjugacy.

2 PMF and likelihood as slices of one surface

The likelihood is not *computed* from the PMF by applying an operation; it *is* the PMF, with the roles of argument and parameter exchanged. "Same functional form" is not a derivation and not a coincidence — it is the definition of the likelihood function.

Fix n and consider the single bivariate function

$$g(v,q) = \binom{n}{v} q^v (1-q)^{n-v}, \qquad (v,q) \in \{0,1,\ldots,n\} \times [0,1]. \tag{6}$$

The PMF and the likelihood are two orthogonal slices of this one surface:
- Fix q, let v vary over $\{0,\ldots,n\}$: this is the PMF $f_Y(v; q, n)$, a slice along the discrete axis.
- Fix the observed v, let q vary over $[0,1]$: this is $L(q \mid v, n)$, a slice along the continuous axis.

So the "computation" is only: substitute the observed v and read what remains as a function of q. The binomial coefficient and both power terms stay algebraically identical, which is why the two expressions are typographically the same. The discrete-versus-continuous distinction is entirely about which axis was left free: v ranges over a finite set, q over a continuum.

Pointwise equality is not identity of objects. The two slices normalize on different axes. Across the discrete axis the PMF sums to 1 for every q, by the binomial theorem:

$$\sum_{v=0}^{n} g(v,q) = \big(q + (1-q)\big)^n = 1. \tag{7}$$

Across the continuous axis the likelihood integrates to

$$\int_0^1 g(v,q)\, dq = \binom{n}{v} \mathrm{B}(v+1,\, n-v+1) = \binom{n}{v} \frac{v!\,(n-v)!}{(n+1)!} = \frac{1}{n+1}, \tag{8}$$

independent of v, and in particular $\neq 1$. Thus L is not a density in q; this is exactly why maximizing it, rather than integrating it, is the sensible operation, and why turning it into a density requires the Beta normalization.

Remark. That the integral (8) equals $\frac{1}{n+1}$ for every v is the statement that a uniform prior on q induces a uniform prior-predictive on the count $v \in \{0,\ldots,n\}$ — the $\mathrm{Beta}(1,1)$–Binomial edge case.

3 The likelihood function, formally

Definition 3.1 (Likelihood function)**.** Let $\{P_\theta : \theta \in \Theta\}$ be a statistical model, a family of probability measures on a sample space $(\mathcal{X}, \mathcal{A})$, all dominated by a common σ-finite measure μ (counting measure in the discrete case, Lebesgue measure in the continuous case). By the Radon–Nikodym theorem each P_θ has a density

$$f(x \mid \theta) = \frac{dP_\theta}{d\mu}(x). \tag{9}$$

Given an observed outcome x, the *likelihood function* is the map

$$L(\cdot \mid x) : \Theta \to [0, \infty), \qquad L(\theta \mid x) = f(x \mid \theta). \tag{10}$$

The single content-bearing move is the exchange of roles from Section 2: it is the same function $f(x \mid \theta)$, but x is held fixed (the data observed) and θ is the free variable. The likelihood is a function on the parameter space, not on the sample space.

Three points that the informal "density read backwards" phrasing obscures:

1. **Defined only up to a θ-independent positive factor.** The Radon–Nikodym derivative depends on the dominating measure: switching $\mu \to \tilde{\mu}$ with $\frac{d\mu}{d\tilde{\mu}} = h(x)$ multiplies every density, hence $L(\theta \mid x)$, by the constant-in-θ factor $h(x)$. This changes nothing inferential — the $\arg\max$, the likelihood ratios $L(\theta_1 \mid x)/L(\theta_2 \mid x)$, and the score $\nabla_\theta \log L$ are all invariant. The likelihood is properly an equivalence class under multiplication by positive functions of x alone. This is why the $\binom{n}{\nu}$ in Section 1 was inert.

2. **$L(\theta \mid x)$ is not a density in θ.** There is no reference measure on Θ in the definition, and $\int_\Theta L(\theta \mid x)\, d\theta$ need not be finite, let alone 1 (the $\frac{1}{n+1}$ computation of (8)). The likelihood becomes a probabilistic object on Θ only after a prior is supplied and the product normalized (Bayes).

3. **The fixed x must lie in the support.** Where all P_θ assign zero μ-density the derivative is defined only μ-a.e., so L is genuinely ambiguous there — harmless, since one evaluates at observed data.

The iid case. For iid data $x = (x_1, \ldots, x_m)$ from $f(\cdot \mid \theta)$, the product structure of the joint density gives

$$L(\theta \mid x) = \prod_{i=1}^{m} f(x_i \mid \theta), \tag{11}$$

and one works with the log-likelihood

$$\ell(\theta) = \log L(\theta \mid x) = \sum_{i=1}^{m} \log f(x_i \mid \theta), \tag{12}$$

the MLE being any $\hat{\theta} \in \arg\max_{\theta \in \Theta} \ell(\theta)$.